

Module 6: Coupled Two-Dimensional Multiphysics Integration

Derivation-Focused Physics Note for RadiationEffects_proj2

Self-consistent coupling of defect evolution, electrostatics, thermal transport, and carrier transport

Purpose of this note. Module 6 is the integration layer of RadiationEffects_proj2. Modules 1–5 each describe one part of the radiation-to-device-degradation chain. Module 6 connects them into one coupled device model. The central idea is that the defect map, temperature map, electrostatic potential map, and carrier map are not independent post-processing products. They feed back on one another through space charge, mobility, recombination, heat generation, defect kinetics, and field-driven transport. This note defines the coupled state variables, writes the governing equations in a common two-dimensional notation, derives the main feedback loops, develops the finite-element discretization structure used for coupling, and gives a practical staggered solver strategy for the first MATLAB implementation.

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Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
2D	Two-dimensional	Reduced project model resolving the in-plane device cross section in coordinates x and y .
3D	Three-dimensional	Full transport or device geometry with coordinates x , y , and z .
BTE	Boltzmann transport equation	Kinetic equation used in Module 4 to motivate ballistic-diffusive heat-carrier transport.
DD	Drift-diffusion	Continuum carrier transport model used in Module 5.
FEM	Finite element method	Spatial discretization method intended for the coupled two-dimensional project model.
PDE	Partial differential equation	Equation involving derivatives with respect to space and/or time.
DAE	Differential-algebraic equation	Coupled system containing both time-evolution equations and constraint equations such as Poisson's equation.
SRH	Shockley-Read-Hall	Defect-assisted recombination mechanism mediated by trap levels inside the band gap.
SEU	Single-event upset	Device-level error mode caused by radiation-induced charge deposition and collection.
TID	Total ionizing dose	Cumulative ionizing-radiation damage metric.
DDD	Displacement damage dose	Damage metric associated with non-ionizing energy loss and lattice displacement.
x, y	In-plane coordinates	Spatial coordinates of the two-dimensional device cross section.
z	Out-of-plane coordinate	Direction averaged out when mapping the three-dimensional Module 1 deposition field into the reduced two-dimensional model.
t	Time	Independent time coordinate.
Ω	Device domain	Two-dimensional material region occupied by the simulated device cross section.
$\partial\Omega$	Device boundary	Boundary of the two-dimensional simulation domain.
$\mathbf{n}$	Unit normal	Outward unit normal vector on $\partial\Omega$.
$\mathbf{U}$	Coupled state vector	Collection of unknown fields, for example $\mathbf{U} = [C_1, \dots, C_N, \phi, T, n, p]^T$.
$\mathbf{C}$	Defect-state vector	Vector of defect concentrations $\mathbf{C} = [C_1, \dots, C_N]^T$.

Symbol / acronym	Name	Meaning in this document
C_j	Defect concentration	Concentration of defect species j , inherited from Module 3.
N	Number of defect species	Number of defect populations retained in the reduced model.
z_j	Defect charge number	Effective integer or averaged charge state of defect species j .
q	Elementary charge magnitude	Positive scalar charge magnitude; electron charge is $-q$, hole charge is $+q$.
ρ	Space-charge density	Charge density entering Poisson's equation.
ρ_{dop}	Dopant charge density	Fixed ionized-dopant contribution to space charge.
ρ_{car}	Carrier charge density	Mobile-carrier contribution $q(p - n)$.
ρ_{def}	Defect charge density	Charged-defect contribution $q \sum_j z_j C_j$.
N_D^+	Ionized donor concentration	Positive fixed donor concentration.
N_A^-	Ionized acceptor concentration	Negative fixed acceptor concentration.
ϕ	Electrostatic potential	Scalar potential solved in Module 2 and reused in Modules 5 and 6.
$\mathbf{E}$	Electric field	Electrostatic field $\mathbf{E} = -\nabla_{xy}\phi$.
ϵ_s	Silicon permittivity	Electrical permittivity of silicon or the local material.
T	Temperature	Lattice or effective thermal field supplied by Module 4.
T_0	Reference temperature	Ambient or initial reference temperature.
ρ_m	Mass density	Material mass density.
$\times$	Specific heat	Heat capacity per unit mass at constant pressure.
k	Thermal conductivity	Diffusive thermal conductivity or effective thermal conductivity.
$\mathbf{q}_{\text{th}}$	Heat flux	Thermal energy flux vector.
$\dot{q}_{\text{rad}}$	Radiation energy-deposition rate	Volumetric radiation energy source from Module 1.
$\dot{q}_{\text{th}}$	Thermalized radiation heat source	Portion of $\dot{q}_{\text{rad}}$ that appears as heat.
Q_J	Joule-heating source	Heat source from electrical power dissipation, often $Q_J = \mathbf{J} \cdot \mathbf{E}$.

Symbol / acronym	Name	Meaning in this document
Q_{rec}	Recombination heat source	Heat released through electron-hole recombination.
Q_{def}	Defect-reaction heat source	Heat associated with defect reactions, annealing, or clustering.
n	Electron concentration	Mobile conduction-band electron density.
p	Hole concentration	Mobile valence-band hole density.
$\mathbf{J}_n$	Electron current density	Conventional current density associated with electron transport.
$\mathbf{J}_p$	Hole current density	Conventional current density associated with hole transport.
$\mathbf{J}$	Total current density	$\mathbf{J} = \mathbf{J}_n + \mathbf{J}_p$.
μ_n	Electron mobility	Electron mobility entering drift-diffusion current.
μ_p	Hole mobility	Hole mobility entering drift-diffusion current.
D_n	Electron diffusivity	Electron diffusion coefficient.
D_p	Hole diffusivity	Hole diffusion coefficient.
D_j	Defect diffusivity	Diffusion coefficient of defect species j .
μ_j	Defect mobility	Field-driven mobility of charged defect species j .
S_j	Defect source	Source term for defect species j , including radiation-induced generation.
R_j	Defect reaction term	Net production or loss of defect species j through reactions, annealing, recombination, or clustering.
G	Carrier generation rate	Volumetric generation rate of electron-hole pairs.
R	Carrier recombination rate	Volumetric removal rate of electron-hole pairs.
R_{SRH}	SRH recombination rate	Defect-assisted recombination rate through trap states.
N_t	Trap density	Concentration of electrically active trap states.
η_{eh}	Carrier-generation efficiency	Fraction of deposited radiation energy assigned to mobile electron-hole pair creation.
ε_{eh}	Pair-creation energy	Mean energy required to create one mobile electron-hole pair.

Symbol / acronym	Name	Meaning in this document
$\mathbf{F}$	Coupled residual vector	Residual form of the monolithic coupled multiphysics system.
$\mathbf{R}$	Discrete residual vector	Residual after spatial and temporal discretization.
Δt	Time step	Discrete time increment.
m	Time-step index	Superscript index for discrete time level t^m .
ℓ	Coupling-iteration index	Superscript index for inner Picard or fixed-point iteration.
ω	Under-relaxation factor	Scalar between 0 and 1 used to stabilize staggered coupling iterations.
ϵ_{coup}	Coupling tolerance	Stopping tolerance for the coupled fixed-point iteration.
L	Characteristic length	Representative device or gradient length scale.
τ	Characteristic time	Representative time scale for a physical process.
Da	Damkohler number	Ratio of transport time scale to reaction time scale.
Pe	Peclet number	Ratio of drift or advective transport to diffusive transport.
Bi	Biot number	Ratio of boundary heat-transfer resistance to internal conduction resistance.
Kn	Knudsen number	Ratio of heat-carrier mean free path to device length scale.
Λ	Mean free path	Average ballistic transport length between scattering events.
y_{device}	Device response	Reduced measurable output such as leakage current, threshold shift, collected charge, or upset probability.
I	Terminal current	Current through a device contact.
V	Terminal voltage	Applied or measured voltage at a device terminal.
$\mathcal{T}_h$	Finite-element mesh	Triangulation of the two-dimensional device domain.
e	Element index	Index for one triangular finite element.
N_a	Shape function	Linear finite-element basis function associated with local or global node a .

Symbol / acronym	Name	Meaning in this document
A_e	Element area	Area of triangular element e .
Φ	Nodal potential vector	Vector containing finite-element nodal values of ϕ .
Θ	Nodal temperature vector	Vector containing finite-element nodal values of T .
$\mathbf{c}_j$	Nodal defect vector	Vector containing nodal values of defect species C_j .
$\mathbf{n}, \mathbf{p}$	Nodal carrier vectors	Vectors containing nodal electron and hole concentrations.
M	Mass matrix	Finite-element matrix from integrals of products of shape functions.
K	Stiffness matrix	Finite-element matrix from gradient-gradient or transport terms.
B_e	Element gradient matrix	Matrix containing gradients of linear triangular shape functions for element e .
$\mathbf{b}$	Source vector	Finite-element load vector assembled from volume and boundary source terms.
$\mathbf{r}$	Discrete residual	Algebraic residual vector used to test convergence of a coupled solve.
$\mathbf{P}_{a \leftarrow b}$	Projection operator	Interpolation or transfer matrix mapping a field from mesh b to mesh a .

1 Role of Module 6 in the project

Modules 1–5 can be run as separate physics blocks. That is useful for understanding each process. It is not enough for a device model, because the fields generated by one module modify the equations solved by the others. Module 6 is therefore the self-consistent integration layer. The reduced project state can be written as

$$\mathbf{U}(x, y, t) = \begin{bmatrix} C_1(x, y, t) \\ \vdots \\ C_N(x, y, t) \\ \phi(x, y, t) \\ T(x, y, t) \\ n(x, y, t) \\ p(x, y, t) \end{bmatrix}. \quad (1.1)$$

Here C_j are defect concentrations, ϕ is electrostatic potential, T is temperature, and n, p are mobile electron and hole concentrations. The corresponding physical maps are

$$\boxed{\text{defect map } \mathbf{C}, \quad \text{potential map } \phi, \quad \text{temperature map } T, \quad \text{carrier map } (n, p).} \quad (1.2)$$

The central message of Module 6 is that each map changes the others:

$$\mathbf{C} \longrightarrow \rho_{\text{def}} \longrightarrow \phi, \mathbf{E}, \quad (1.3)$$

$$\mathbf{C} \longrightarrow \mu_n, \mu_p, R_{\text{SRH}} \longrightarrow n, p, \mathbf{J}, \quad (1.4)$$

$$T \longrightarrow D_j, R_j, \mu_n, \mu_p, R_{\text{SRH}}, \quad (1.5)$$

$$\phi, \mathbf{E} \longrightarrow \text{carrier drift and charged-defect drift}, \quad (1.6)$$

$$n, p \longrightarrow \rho_{\text{car}} \longrightarrow \phi, \mathbf{E}, \quad (1.7)$$

$$\mathbf{J}, R, R_j \longrightarrow \text{heat generation} \longrightarrow T. \quad (1.8)$$

Core intuition. Module 6 is not a new isolated physical mechanism. It is the statement that radiation damage, heat, electric field, and charge transport form a feedback system. A defect map changes space charge and recombination. Space charge changes the electric field. The electric field changes carrier motion and sometimes defect motion. Carrier motion and recombination generate heat. Heat changes defect kinetics and carrier mobility. The device response emerges from the fixed point of these interactions.

2 The four coupled maps over the same device cross section

The phrase “coupled maps” means that the same material point (x, y) carries multiple fields. At a location near a radiation-damaged region, one may simultaneously have a high defect density C_j , altered space charge ρ , a distorted potential ϕ , elevated temperature T , reduced mobility μ_n, μ_p , and increased recombination R . These are not separate pictures of separate devices; they are different projections of the same physical state.

2.1 Defect map

The defect map is the set of species concentrations

$$\mathbf{C}(x, y, t) = [C_1(x, y, t), C_2(x, y, t), \dots, C_N(x, y, t)]^T. \quad (2.1)$$

Examples include vacancies, interstitials, vacancy-oxygen complexes, divacancies, charged traps, or effective reduced defect populations. The Module 3 equations evolve C_j through diffusion, drift, reactions, annealing, and sources produced by radiation deposition.

2.2 Potential map

The potential map is

$$\phi(x, y, t), \quad \mathbf{E}(x, y, t) = -\nabla_{xy}\phi(x, y, t). \quad (2.2)$$

It comes from the charge density produced by ionized dopants, mobile carriers, and charged defects. In Module 6, ϕ is not merely a fixed bias field; it responds to radiation-induced changes in $\mathbf{C}$, n , and p .

2.3 Temperature map

The temperature map is

$$T(x, y, t). \quad (2.3)$$

It is driven by the thermalized fraction of radiation energy deposition, electrical power dissipation, recombination, and defect reactions. It feeds back into almost every coefficient: defect diffusivity, annealing rates, carrier mobility, intrinsic concentration, and recombination lifetimes.

2.4 Carrier map

The carrier map is

$$(n(x, y, t), p(x, y, t)). \quad (2.4)$$

It determines the mobile charge density, current density, recombination rate, and part of the heat generation. Module 5 computes these fields using drift-diffusion equations in the electrostatic, thermal, and defect environment supplied by the other modules.

3 Coupled governing equations

This section writes the four map equations in a common notation. These equations are more general than the first implementation will need. The first implementation can turn off selected feedbacks while preserving the same architecture.

3.1 Defect evolution equation

For defect species j , a general two-dimensional conservation form is

$$\frac{\partial C_j}{\partial t} = -\nabla_{xy} \cdot \mathbf{\Gamma}_j + R_j(\mathbf{C}, T, \mathbf{E}, n, p) + S_j(\dot{q}_{\text{rad}}, T), \quad (3.1)$$

where $\mathbf{\Gamma}_j$ is the defect particle flux, R_j is the net reaction term, and S_j is the radiation-induced source term. A useful reduced flux model is

$$\mathbf{\Gamma}_j = -D_j(T, \mathbf{E}, \mathbf{C}) \nabla_{xy} C_j + \mu_j z_j q C_j \mathbf{E}. \quad (3.2)$$

The first term is diffusion down concentration gradients. The second term is drift of charged defects under the electric field. The sign is controlled by the charge number z_j . A positively charged defect drifts along $\mathbf{E}$, while a negatively charged defect drifts opposite $\mathbf{E}$.

Substituting Eq. (3.2) into Eq. (3.1) gives

$$\boxed{\frac{\partial C_j}{\partial t} = \nabla_{xy} \cdot (D_j \nabla_{xy} C_j) - \nabla_{xy} \cdot (\mu_j z_j q C_j \mathbf{E}) + R_j + S_j.} \quad (3.3)$$

This is the Module 3 equation embedded inside the Module 6 feedback loop.

3.2 Electrostatic potential equation

The electrostatic equation is Poisson's equation,

$$\boxed{\nabla_{xy} \cdot (\epsilon_s \nabla_{xy} \phi) = -\rho, \quad \mathbf{E} = -\nabla_{xy} \phi.} \quad (3.4)$$

The total space-charge density is decomposed as

$$\rho = \rho_{\text{dop}} + \rho_{\text{car}} + \rho_{\text{def}}. \quad (3.5)$$

For the silicon regions of the reduced model,

$$\rho_{\text{dop}} = q (N_D^+ - N_A^-), \quad (3.6)$$

$$\rho_{\text{car}} = q (p - n), \quad (3.7)$$

$$\rho_{\text{def}} = q \sum_{j=1}^N z_j C_j. \quad (3.8)$$

Therefore

$$\boxed{\rho = q \left[p - n + N_D^+ - N_A^- + \sum_{j=1}^N z_j C_j \right].} \quad (3.9)$$

Equation (3.9) is one of the most important coupling equations in the project. It states that radiation-induced charged defects enter electrostatics in the same charge-density budget as dopants and mobile carriers.

3.3 Thermal equation

A diffusive thermal balance can be written as

$$\boxed{\rho_m c_p \frac{\partial T}{\partial t} = -\nabla_{xy} \cdot \mathbf{q}_{th} + Q_{tot}, \quad \mathbf{q}_{th} = -k \nabla_{xy} T.} \quad (3.10)$$

In the simplest Fourier form this becomes

$$\boxed{\rho_m c_p \frac{\partial T}{\partial t} = \nabla_{xy} \cdot (k \nabla_{xy} T) + Q_{tot}.} \quad (3.11)$$

Module 4 uses a more detailed ballistic-diffusive thermal model, but the coupling role is the same: Module 6 requires a temperature field and a thermal-energy balance.

The total heat source can be decomposed as

$$\boxed{Q_{tot} = \dot{q}_{th} + Q_J + Q_{rec} + Q_{def} + Q_{ext}.} \quad (3.12)$$

Here $\dot{q}_{th}$ is the thermalized part of radiation energy deposition, Q_J is Joule heating, Q_{rec} is recombination heat, Q_{def} is heat from defect reactions or annealing, and Q_{ext} is any externally imposed thermal source.

A common reduced form for Joule heating is

$$\boxed{Q_J = \mathbf{J} \cdot \mathbf{E} = (\mathbf{J}_n + \mathbf{J}_p) \cdot \mathbf{E}.} \quad (3.13)$$

This term couples carrier transport directly back into the temperature map.

3.4 Carrier transport equations

Module 5 supplies the electron and hole continuity equations:

$$\boxed{\frac{\partial n}{\partial t} = \frac{1}{q} \nabla_{xy} \cdot \mathbf{J}_n + G - R,} \quad (3.14)$$

$$\boxed{\frac{\partial p}{\partial t} = -\frac{1}{q} \nabla_{xy} \cdot \mathbf{J}_p + G - R.} \quad (3.15)$$

The drift-diffusion current densities are

$$\mathbf{J}_n = q \mu_n n \mathbf{E} + q D_n \nabla_{xy} n, \quad (3.16)$$

$$\mathbf{J}_p = q \mu_p p \mathbf{E} - q D_p \nabla_{xy} p. \quad (3.17)$$

The signs in Eqs. (3.16)–(3.17) are conventional-current signs. Electron particles drift opposite the electric field, but electron conventional current points along the electric field.

Radiation can generate mobile electron-hole pairs through

$$G_{\text{rad}}(x, y, t) = \frac{\eta_{\text{eh}} \dot{q}_{\text{rad}}(x, y, t)}{\varepsilon_{\text{eh}}}. \quad (3.18)$$

The total generation rate may include other terms,

$$G = G_{\text{rad}} + G_{\text{th}} + G_{\text{inj}}, \quad (3.19)$$

where G_{th} is thermal generation and G_{inj} is effective carrier injection.

Defect-assisted recombination can be represented by an SRH term. For one effective trap population, a standard expression is

$$R_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}. \quad (3.20)$$

In Module 6, the trap density is linked to the defect map, for example

$$N_t(x, y, t) = \sum_{j=1}^N \eta_j C_j(x, y, t), \quad (3.21)$$

so the recombination lifetimes depend on radiation-induced defects:

$$\tau_n = \frac{1}{\sigma_n v_{\text{th},n} N_t}, \quad \tau_p = \frac{1}{\sigma_p v_{\text{th},p} N_t}. \quad (3.22)$$

4 Where the feedback loops come from

4.1 Defects perturb electrostatics

The direct coupling from defects to electrostatics follows from Eq. (3.9). If defect species j has charge number z_j , it contributes

$$\rho_{\text{def},j} = q z_j C_j. \quad (4.1)$$

The change in charge density caused by a defect perturbation δC_j is

$$\delta \rho = q z_j \delta C_j. \quad (4.2)$$

Poisson's equation then gives the potential perturbation

$$\nabla_{xy} \cdot (\varepsilon_s \nabla_{xy} \delta \phi) = -q z_j \delta C_j. \quad (4.3)$$

Thus a localized charged-defect cloud bends the electrostatic field. If the charged defect lies in or near a depletion region, the effect can alter junction fields, leakage, and charge-collection paths.

4.2 Electrostatics steers carriers and charged defects

The electric field enters carrier transport through the drift terms in Eqs. (3.16)–(3.17). It can also enter charged-defect transport through Eq. (3.2). Therefore the same potential map can steer both fast electronic transport and slower defect redistribution. This is why a damaged device cannot always be modeled by solving defect evolution first and electrostatics only once afterward. If the defect charge strongly alters $\mathbf{E}$, then $\mathbf{E}$ can in turn change later defect drift or carrier collection.

4.3 Defects modify mobility

Defects scatter carriers and shorten the momentum-relaxation time. A reduced mobility model can be written as

$$\mu_n(T, \mathbf{C}) = \frac{\mu_{n0}(T)}{1 + \sum_j \alpha_{n,j} C_j}, \quad (4.4)$$

$$\mu_p(T, \mathbf{C}) = \frac{\mu_{p0}(T)}{1 + \sum_j \alpha_{p,j} C_j}. \quad (4.5)$$

This is not the only possible model, but it captures the intended physics: increasing defect concentration lowers the effective mobility, and therefore lowers drift current for a given field and carrier density.

4.4 Defects modify recombination

Defects also produce trap levels. Using Eqs. (3.21)–(3.22), an increase in C_j raises N_t , lowers τ_n and τ_p , and usually increases SRH recombination. In the carrier equations, this removes mobile electrons and holes:

$$G - R \quad \text{decreases as} \quad R_{\text{SRH}} \text{ increases.} \quad (4.6)$$

That reduced carrier population then feeds back into space charge through $q(p - n)$ and into current through $\mathbf{J}_n + \mathbf{J}_p$.

4.5 Temperature modifies almost every coefficient

Temperature enters defect kinetics through Arrhenius-type diffusivities and rates. A typical defect diffusivity is

$$D_j(T) = D_{j0} \exp\left(-\frac{E_{D,j}}{k_B T}\right), \quad (4.7)$$

and a typical annealing coefficient is

$$k_{\text{ann},j}(T) = k_{\text{ann},j0} \exp\left(-\frac{E_{A,j}}{k_B T}\right). \quad (4.8)$$

Temperature also changes carrier mobility and intrinsic carrier concentration. A common reduced mobility scaling is

$$\mu_{n0}(T) = \mu_{n,\text{ref}} \left(\frac{T}{T_{\text{ref}}}\right)^{-\gamma_n}, \quad \mu_{p0}(T) = \mu_{p,\text{ref}} \left(\frac{T}{T_{\text{ref}}}\right)^{-\gamma_p}. \quad (4.9)$$

Thus a local hot spot can simultaneously increase defect mobility, change annealing rates, change carrier mobility, increase thermal generation, and alter recombination kinetics.

4.6 Carriers and currents generate heat

Carrier transport feeds the thermal equation through Joule heating. Starting from the electrical power density, the local work done by the field on charge carriers is

$$Q_J = \mathbf{J} \cdot \mathbf{E}. \quad (4.10)$$

In a region with high field and high current density, this can increase T , which then changes μ_n , μ_p , D_j , and reaction rates. This is the electrical-to-thermal feedback loop.

Recombination can also release energy. A reduced model can write

$$Q_{\text{rec}} \approx E_{\text{rec}} R, \quad (4.11)$$

where E_{rec} is an effective energy released as heat per recombination event. The exact partition between photons, phonons, and electronic excitation depends on the material and recombination channel. For the first implementation, Q_{rec} can be omitted or treated as a secondary correction unless recombination heating is important.

5 Monolithic coupled system

A compact way to write Module 6 is as a coupled residual equation

$$\boxed{\mathbf{F}\left(\mathbf{U}, \frac{\partial \mathbf{U}}{\partial t}; \dot{q}_{\text{rad}}, \text{BCs, parameters}\right) = \mathbf{0}.} \quad (5.1)$$

Expanded by physics block, the monolithic system contains

$$F_{C_j} \equiv \frac{\partial C_j}{\partial t} - \nabla_{xy} \cdot (D_j \nabla_{xy} C_j) + \nabla_{xy} \cdot (\mu_j z_j q C_j \mathbf{E}) - R_j - S_j = 0, \quad (5.2)$$

$$F_\phi \equiv \nabla_{xy} \cdot (\varepsilon_s \nabla_{xy} \phi) + \rho = 0, \quad (5.3)$$

$$F_T \equiv \rho_m c_p \frac{\partial T}{\partial t} - \nabla_{xy} \cdot (k \nabla_{xy} T) - Q_{\text{tot}} = 0, \quad (5.4)$$

$$F_n \equiv \frac{\partial n}{\partial t} - \frac{1}{q} \nabla_{xy} \cdot \mathbf{J}_n - G + R = 0, \quad (5.5)$$

$$F_p \equiv \frac{\partial p}{\partial t} + \frac{1}{q} \nabla_{xy} \cdot \mathbf{J}_p - G + R = 0. \quad (5.6)$$

Equations (5.2)–(5.6) are strongly coupled because coefficients and source terms inside each residual depend on the full state $\mathbf{U}$. For example,

$$D_j = D_j(T, \mathbf{E}, \mathbf{C}), \quad (5.7)$$

$$\mu_n = \mu_n(T, \mathbf{C}), \quad (5.8)$$

$$R = R(n, p, T, \mathbf{C}), \quad (5.9)$$

$$\rho = \rho(n, p, \mathbf{C}), \quad (5.10)$$

$$Q_{\text{tot}} = Q_{\text{tot}}(\dot{q}_{\text{rad}}, \mathbf{J}, \mathbf{E}, R, R_j). \quad (5.11)$$

Why this is a DAE-like system. The defect, thermal, and carrier equations are time-evolution equations. Poisson's equation is an instantaneous constraint on ϕ for the current charge distribution. Therefore the coupled model behaves like a PDE/DAE system: some variables evolve dynamically, while ϕ is constrained by the present state.

6 Weak forms for a two-dimensional FEM implementation

This section states the weak forms that connect naturally to finite element implementation. The goal is not to implement all of them immediately, but to make the mathematical structure explicit.

6.1 Poisson weak form

Start from

$$\nabla_{xy} \cdot (\varepsilon_s \nabla_{xy} \phi) = -\rho. \quad (6.1)$$

Multiply by a test function v and integrate over Ω :

$$\int_{\Omega} v \nabla_{xy} \cdot (\varepsilon_s \nabla_{xy} \phi) d\Omega = - \int_{\Omega} v \rho d\Omega. \quad (6.2)$$

Integrating the left side by parts gives

$$- \int_{\Omega} \varepsilon_s \nabla_{xy} v \cdot \nabla_{xy} \phi d\Omega + \int_{\partial\Omega} v \varepsilon_s \nabla_{xy} \phi \cdot \mathbf{n} d\Gamma = - \int_{\Omega} v \rho d\Omega. \quad (6.3)$$

Rearranging,

$$\boxed{\int_{\Omega} \varepsilon_s \nabla_{xy} v \cdot \nabla_{xy} \phi \, d\Omega = \int_{\Omega} v \rho \, d\Omega + \int_{\partial\Omega} v \varepsilon_s \nabla_{xy} \phi \cdot \mathbf{n} \, d\Gamma.} \quad (6.4)$$

Dirichlet contacts set ϕ directly. Neumann or insulating boundaries set the normal displacement field.

6.2 Thermal weak form

For the Fourier form,

$$\rho_m c_p \frac{\partial T}{\partial t} = \nabla_{xy} \cdot (k \nabla_{xy} T) + Q_{\text{tot}}. \quad (6.5)$$

Using test function w , the weak form is

$$\boxed{\int_{\Omega} w \rho_m c_p \frac{\partial T}{\partial t} \, d\Omega + \int_{\Omega} k \nabla_{xy} w \cdot \nabla_{xy} T \, d\Omega = \int_{\Omega} w Q_{\text{tot}} \, d\Omega + \int_{\partial\Omega} w k \nabla_{xy} T \cdot \mathbf{n} \, d\Gamma.} \quad (6.6)$$

For the Module 4 ballistic-diffusive model, this weak form is replaced or augmented by the reduced ballistic-diffusive variables. The coupling logic remains the same: Module 6 requires a temperature update consistent with the total heat source.

6.3 Defect weak form

Starting from

$$\frac{\partial C_j}{\partial t} + \nabla_{xy} \cdot \mathbf{\Gamma}_j = R_j + S_j, \quad (6.7)$$

multiply by a test function s and integrate:

$$\int_{\Omega} s \frac{\partial C_j}{\partial t} \, d\Omega + \int_{\Omega} s \nabla_{xy} \cdot \mathbf{\Gamma}_j \, d\Omega = \int_{\Omega} s (R_j + S_j) \, d\Omega. \quad (6.8)$$

After integration by parts,

$$\boxed{\int_{\Omega} s \frac{\partial C_j}{\partial t} \, d\Omega - \int_{\Omega} \nabla_{xy} s \cdot \mathbf{\Gamma}_j \, d\Omega + \int_{\partial\Omega} s \mathbf{\Gamma}_j \cdot \mathbf{n} \, d\Gamma = \int_{\Omega} s (R_j + S_j) \, d\Omega.} \quad (6.9)$$

This form is useful because sinks, reflective boundaries, and interface loss terms can all be expressed as boundary fluxes.

6.4 Carrier weak forms

For electrons,

$$\frac{\partial n}{\partial t} - \frac{1}{q} \nabla_{xy} \cdot \mathbf{J}_n = G - R. \quad (6.10)$$

Multiplying by a test function a , integrating, and applying integration by parts gives

$$\boxed{\int_{\Omega} a \frac{\partial n}{\partial t} d\Omega + \frac{1}{q} \int_{\Omega} \nabla_{xy} a \cdot \mathbf{J}_n d\Omega - \frac{1}{q} \int_{\partial\Omega} a \mathbf{J}_n \cdot \mathbf{n} d\Gamma = \int_{\Omega} a(G - R) d\Omega.} \quad (6.11)$$

For holes,

$$\frac{\partial p}{\partial t} + \frac{1}{q} \nabla_{xy} \cdot \mathbf{J}_p = G - R, \quad (6.12)$$

so

$$\boxed{\int_{\Omega} b \frac{\partial p}{\partial t} d\Omega - \frac{1}{q} \int_{\Omega} \nabla_{xy} b \cdot \mathbf{J}_p d\Omega + \frac{1}{q} \int_{\partial\Omega} b \mathbf{J}_p \cdot \mathbf{n} d\Gamma = \int_{\Omega} b(G - R) d\Omega.} \quad (6.13)$$

The opposite signs follow from the sign difference between electron conventional current and hole conventional current in the continuity equations.

7 Finite-element discretization of the coupled Module 6 system

The previous section gave the weak forms. This section makes the finite-element implementation explicit. The first Module 6 MATLAB path uses the same linear triangular mesh style developed for Modules 2–5. This choice is deliberate: the goal is not only to solve each physics block, but to pass fields between blocks without ambiguous interpolation or unit conversion.

7.1 Shared mesh and nodal unknowns

Let $\mathcal{T}_h$ be a triangular finite-element mesh of the two-dimensional device domain Ω . The domain is approximated as

$$\Omega \approx \Omega_h = \bigcup_{e=1}^{N_e} \Omega_e, \quad (7.1)$$

where Ω_e is one triangular element and N_e is the number of elements. For a three-node linear triangle, each element carries three local shape functions,

$$N_1(x, y), \quad N_2(x, y), \quad N_3(x, y), \quad (7.2)$$

with the interpolation property

$$N_a(\text{node } b) = \delta_{ab}. \quad (7.3)$$

For a coupled Module 6 solve, the finite-element approximations are

$$C_j^h(x, y, t) = \sum_{a=1}^{N_n} N_a(x, y) c_{j,a}(t), \quad (7.4)$$

$$\phi^h(x, y, t) = \sum_{a=1}^{N_n} N_a(x, y) \Phi_a(t), \quad (7.5)$$

$$T^h(x, y, t) = \sum_{a=1}^{N_n} N_a(x, y) \Theta_a(t), \quad (7.6)$$

$$n^h(x, y, t) = \sum_{a=1}^{N_n} N_a(x, y) n_a(t), \quad (7.7)$$

$$p^h(x, y, t) = \sum_{a=1}^{N_n} N_a(x, y) p_a(t). \quad (7.8)$$

Here N_n is the number of global mesh nodes. The unknowns are the nodal coefficients $c_{j,a}$, Φ_a , Θ_a , n_a , and p_a . In vector form,

$$\widetilde{\mathbf{U}} = \begin{bmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_N \\ \Phi \\ \Theta \\ \mathbf{n} \\ \mathbf{p} \end{bmatrix}. \quad (7.9)$$

This is the discrete counterpart of the continuum state vector in Eq. (1.1).

Why a shared mesh matters. If Modules 2–5 use different meshes, every coupling term requires interpolation. That is possible, but it creates another source of error. The first Module 6 FEM implementation therefore uses one shared triangular mesh so that C_j , ϕ , T , n , and p are all stored at the same nodes. This makes the first coupling path easier to verify.

7.2 Element-level matrices on a linear triangle

For a linear triangular element, the gradients of the shape functions are constant inside the element. If the local nodal values of a scalar field u are stored in $\mathbf{u}_e = [u_1, u_2, u_3]^T$, then

$$\nabla_{xy} u^h|_{\Omega_e} = B_e \mathbf{u}_e, \quad (7.10)$$

where B_e is the element gradient matrix. This constant-gradient property makes the basic element matrices especially transparent.

The scalar mass matrix is

$$M_{ab}^{(e)} = \int_{\Omega_e} N_a N_b d\Omega = \frac{A_e}{12} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}_{ab}. \quad (7.11)$$

The diffusion or conductivity stiffness matrix for coefficient κ is

$$K_{ab}^{(e)} = \int_{\Omega_e} \kappa \nabla_{xy} N_a \cdot \nabla_{xy} N_b d\Omega = \kappa_e A_e \nabla_{xy} N_a \cdot \nabla_{xy} N_b. \quad (7.12)$$

The source vector for a nodal source $s^h = \sum_b N_b s_b$ is

$$b_a^{(e)} = \int_{\Omega_e} N_a s^h d\Omega = \sum_{b=1}^3 M_{ab}^{(e)} s_b. \quad (7.13)$$

Equations (7.11)–(7.13) are reused across defect diffusion, heat conduction, Poisson electrostatics, and carrier diffusion. The physics changes through the coefficient κ , the source term, the sign convention, and the variable being solved.

7.3 Discrete Poisson block

The Poisson weak form in Eq. (6.4) gives the algebraic system

$$K_\phi \Phi = \mathbf{b}_\rho + \mathbf{b}_{\Gamma, \phi}. \quad (7.14)$$

For a uniform silicon permittivity, the element matrix is

$$(K_\phi^{(e)})_{ab} = \int_{\Omega_e} \varepsilon_s \nabla_{xy} N_a \cdot \nabla_{xy} N_b d\Omega. \quad (7.15)$$

The source vector is assembled from

$$(b_\rho^{(e)})_a = \int_{\Omega_e} N_a \rho^h d\Omega, \quad (7.16)$$

where the nodal charge density is

$$\rho_a = q \left[p_a - n_a + N_{D,a}^+ - N_{A,a}^- + \sum_{j=1}^N z_j c_{j,a} \right]. \quad (7.17)$$

Dirichlet contact potentials are imposed strongly by modifying rows and columns of the global matrix. Zero-normal-displacement boundaries are natural boundary conditions and require no extra source term.

7.4 Discrete defect block

For one defect species without field drift, the FEM semi-discrete equation is

$$M_C \dot{\mathbf{c}}_j + (K_{D,j} + K_{R,j}) \mathbf{c}_j = \mathbf{s}_j. \quad (7.18)$$

Here

$$(M_C^{(e)})_{ab} = \int_{\Omega_e} N_a N_b d\Omega, \quad (7.19)$$

$$(K_{D,j}^{(e)})_{ab} = \int_{\Omega_e} D_j \nabla_{xy} N_a \cdot \nabla_{xy} N_b d\Omega, \quad (7.20)$$

$$(K_{R,j}^{(e)})_{ab} = \int_{\Omega_e} k_{\text{ann},j} N_a N_b d\Omega. \quad (7.21)$$

A backward-Euler update gives

$$\boxed{[M_C + \Delta t (K_{D,j} + K_{R,j})] \mathbf{c}_j^{m+1} = M_C \mathbf{c}_j^m + \Delta t \mathbf{s}_j^{m+1}.} \quad (7.22)$$

If charged-defect drift is retained, an additional nonsymmetric drift matrix appears because the flux contains $\mu_j z_j q C_j \mathbf{E}$. In the first coupled MATLAB path, this term is kept as a documented extension because the most important first coupling is the map $C_j \rightarrow \rho_{\text{def}} \rightarrow \phi$.

7.5 Discrete thermal block

For the Fourier-limit thermal update used inside the first Module 6 coupling scaffold,

$$M_T \dot{\Theta} + K_T \Theta = \mathbf{q}_{\text{tot}}, \quad (7.23)$$

where

$$(M_T^{(e)})_{ab} = \int_{\Omega_e} \rho_m c_p N_a N_b d\Omega, \quad (7.24)$$

$$(K_T^{(e)})_{ab} = \int_{\Omega_e} k \nabla_{xy} N_a \cdot \nabla_{xy} N_b d\Omega. \quad (7.25)$$

The backward-Euler update is

$$\boxed{\left(\frac{1}{\Delta t} M_T + K_T \right) \Theta^{m+1} = \frac{1}{\Delta t} M_T \Theta^m + \mathbf{q}_{\text{tot}}^{m+1}.} \quad (7.26)$$

The full Module 4 ballistic-diffusive FEM path can replace this thermal sub-block. The reduced Fourier-limit form is still useful in Module 6 because it tests the coupling of heat sources, temperature-dependent coefficients, and FEM field transfer before adding the second-order ballistic-diffusive thermal state.

7.6 Discrete carrier blocks

For an electron-like scalar carrier variable, the reduced FEM form can be written as

$$\boxed{M_n \dot{\mathbf{n}} + (K_{D,n} + K_{E,n} + K_{R,n}) \mathbf{n} = \mathbf{g}_n + \mathbf{r}_{n,\text{eq}}.} \quad (7.27)$$

The hole block has the same structure but the drift sign changes. The diffusion matrix is built from $D_n \nabla_{xy} N_a \cdot \nabla_{xy} N_b$, while the drift matrix comes from the weak form terms involving $\mu_n \mathbf{E} n$. With backward Euler,

$$\boxed{\left[\frac{1}{\Delta t} M_n + K_{D,n} + K_{E,n} + K_{R,n} \right] \mathbf{n}^{m+1} = \frac{1}{\Delta t} M_n \mathbf{n}^m + \mathbf{g}_n + \mathbf{r}_{n,\text{eq}}.} \quad (7.28)$$

An analogous equation is solved for $\mathbf{p}^{m+1}$. In Module 6 the coefficients inside these matrices depend on the current coupled estimates of T , $\mathbf{E}$, and $\mathbf{C}$:

$$\mu_n = \mu_n(T, \mathbf{C}), \quad D_n = D_n(T, \mu_n), \quad K_{R,n} = K_{R,n}(\mathbf{C}, T). \quad (7.29)$$

7.7 Discrete coupled residual and block structure

If all fields are solved at once, the coupled finite-element equations form one large nonlinear residual,

$$\boxed{\mathbf{R}(\widetilde{\mathbf{U}}^{m+1}; \widetilde{\mathbf{U}}^m) = \mathbf{0}.} \quad (7.30)$$

A schematic block structure is

$$\begin{bmatrix} R_C(\mathbf{c}, \Phi, \Theta, \mathbf{n}, \mathbf{p}) \\ R_\phi(\mathbf{c}, \Phi, \mathbf{n}, \mathbf{p}) \\ R_T(\mathbf{c}, \Phi, \Theta, \mathbf{n}, \mathbf{p}) \\ R_n(\mathbf{c}, \Phi, \Theta, \mathbf{n}, \mathbf{p}) \\ R_p(\mathbf{c}, \Phi, \Theta, \mathbf{n}, \mathbf{p}) \end{bmatrix} = \mathbf{0}. \quad (7.31)$$

The off-diagonal dependencies are the multiphysics couplings. For example, R_ϕ depends on $\mathbf{c}$, $\mathbf{n}$, and $\mathbf{p}$ through charge density; R_n depends on Φ through the electric field and on $\mathbf{c}$ through defect-dependent mobility and recombination; R_T depends on Φ , $\mathbf{n}$, and $\mathbf{p}$ through $\mathbf{J} \cdot \mathbf{E}$.

A monolithic Newton method would linearize this entire residual and solve one large block Jacobian. The first MATLAB implementation instead uses a staggered fixed-point method, because each sub-block has already been tested in Modules 2–5 and the coupling terms can be verified one at a time.

7.8 Field transfer and interpolation policy

The first coupled implementation uses one common mesh. If later modules use different meshes, then field transfer must be explicit. If $\mathbf{u}_b$ is a nodal field on mesh b , its projection to mesh a should be written as

$$\mathbf{u}_a = \mathbf{P}_{a \leftarrow b} \mathbf{u}_b. \quad (7.32)$$

Examples include projecting Module 3 defect fields onto the Module 2 electrostatic mesh, or projecting Module 2 electric fields onto the Module 5 transport mesh. These projection operators should be documented and tested because interpolation errors can masquerade as physics.

7.9 MATLAB implementation map for Module 6

The first executable Module 6 FEM path follows this code structure:

- `main_module6_multiphysics.m`: top-level entry point.
- `default_module6_fem_params.m`: domain, coupling, material, defect, carrier, thermal, and boundary parameters.
- `initialize_module6_fem_state.m`: initializes C , ϕ , T , n , and p on the shared mesh.
- `solve_coupled_multiphysics_fem_2d.m`: staggered finite-element coupling loop.
- `update_module6_electrostatics.m`: builds ρ , solves Poisson's equation, and computes $\mathbf{E}$.
- `compute_module6_metrics.m`: evaluates convergence and scalar diagnostics.
- `write_module6_fem_summary.m`: writes a human-readable summary file.

The important conceptual point is that Module 6 does not introduce a new element type. It reuses the same finite-element interpolation and assembly logic from Modules 2–5, then makes the source terms and coefficients depend on the other solved fields.

8 Staggered coupling strategy for the first implementation

A fully monolithic Newton solve for all fields at once is mathematically clean but not the best first implementation. A more practical first path is a staggered fixed-point iteration. At time level t^{m+1} , initialize

$$\mathbf{U}^{m+1,0} = \mathbf{U}^m. \quad (8.1)$$

Then repeat the following steps for coupling iteration $\ell = 0, 1, 2, \dots$:

1. **Defect update.** Solve Module 3 for $\mathbf{C}^{m+1,\ell+1}$ using the current estimates $T^{m+1,\ell}$, $\mathbf{E}^{m+1,\ell}$, and radiation source $\dot{q}_{\text{rad}}$.
2. **Electrostatic update.** Compute $\rho^{m+1,\ell+1}$ from $\mathbf{C}^{m+1,\ell+1}$, $n^{m+1,\ell}$, and $p^{m+1,\ell}$. Solve Poisson's equation for $\phi^{m+1,\ell+1}$, then set $\mathbf{E}^{m+1,\ell+1} = -\nabla_{xy}\phi^{m+1,\ell+1}$.
3. **Carrier update.** Solve Module 5 for $n^{m+1,\ell+1}$ and $p^{m+1,\ell+1}$ using $\mathbf{E}^{m+1,\ell+1}$, $T^{m+1,\ell}$, and $\mathbf{C}^{m+1,\ell+1}$. Compute $\mathbf{J}_n^{m+1,\ell+1}$, $\mathbf{J}_p^{m+1,\ell+1}$, and $R^{m+1,\ell+1}$.
4. **Thermal update.** Compute $Q_{\text{tot}}^{m+1,\ell+1}$ from $\dot{q}_{\text{th}}$, $\mathbf{J} \cdot \mathbf{E}$, recombination, and any defect-reaction heat. Solve Module 4 for $T^{m+1,\ell+1}$.
5. **Relaxation and convergence check.** Optionally apply under-relaxation, then stop when all fields change by less than the coupling tolerance.

Under-relaxation replaces a raw update $U_{\text{raw}}^{\ell+1}$ by

$$U^{\ell+1} = (1 - \omega)U^{\ell} + \omega U_{\text{raw}}^{\ell+1}, \quad 0 < \omega \leq 1. \quad (8.2)$$

A simple convergence metric is

$$\epsilon_{\text{coup}} = \max_{u \in \{C_1, \dots, C_N, \phi, T, n, p\}} \frac{\|u^{\ell+1} - u^{\ell}\|_2}{\|u^{\ell+1}\|_2 + u_{\text{floor}}}. \quad (8.3)$$

Here u_{floor} prevents division by zero when a field is initially very small.

Interpretation of the staggered algorithm. The algorithm is a search for a self-consistent set of maps. The defect map used in electrostatics must match the defect map produced by kinetics. The electric field used in carrier transport must match the field produced by charge density. The temperature used in reaction rates must match the temperature produced by heat sources. If those maps stop changing under repeated cycling, the coupled system has reached a fixed point for that time step.

9 Boundary and initial conditions

9.1 Defect boundary conditions

Defect boundaries can represent reflection, sinks, or interfaces. A no-flux boundary is

$$\mathbf{\Gamma}_j \cdot \mathbf{n} = 0. \quad (9.1)$$

A perfect sink sets the concentration at the interface,

$$C_j = C_{j,\text{sink}}, \quad (9.2)$$

with $C_{j,\text{sink}} = 0$ for an ideal absorber. A finite surface-reaction sink can be written as

$$\mathbf{\Gamma}_j \cdot \mathbf{n} = s_j(C_j - C_{j,\text{eq}}), \quad (9.3)$$

where s_j is an interface sink velocity.

9.2 Electrostatic boundary conditions

Contacts impose Dirichlet potential conditions,

$$\phi = V_a \quad \text{on contact } a. \quad (9.4)$$

Insulating boundaries usually impose a normal electric-displacement condition,

$$\epsilon_s \nabla_{xy} \phi \cdot \mathbf{n} = 0, \quad (9.5)$$

unless surface charge is included.

9.3 Thermal boundary conditions

A fixed-temperature boundary is

$$T = T_b. \quad (9.6)$$

A convective boundary is

$$-k \nabla_{xy} T \cdot \mathbf{n} = h(T - T_\infty), \quad (9.7)$$

where h is the heat-transfer coefficient and T_∞ is the external ambient temperature. For the ballistic-diffusive Module 4 model, boundary emission and re-entry can be represented by effective boundary conditions on heat-carrier populations or by an equivalent reduced thermal boundary term.

9.4 Carrier boundary conditions

Ohmic contacts can prescribe equilibrium or quasi-equilibrium carrier values. Insulating boundaries impose zero normal current,

$$\mathbf{J}_n \cdot \mathbf{n} = 0, \quad \mathbf{J}_p \cdot \mathbf{n} = 0. \quad (9.8)$$

Surface recombination can be modeled by normal current losses proportional to carrier excess:

$$\frac{1}{q} \mathbf{J}_n \cdot \mathbf{n} = S_n(n - n_{\text{eq}}), \quad (9.9)$$

$$-\frac{1}{q} \mathbf{J}_p \cdot \mathbf{n} = S_p(p - p_{\text{eq}}), \quad (9.10)$$

where S_n and S_p are surface recombination velocities.

9.5 Initial conditions

A natural coupled simulation starts from a pre-irradiation equilibrium state:

$$C_j(x, y, 0) = C_{j0}(x, y), \quad (9.11)$$

$$T(x, y, 0) = T_0(x, y), \quad (9.12)$$

$$n(x, y, 0) = n_0(x, y), \quad (9.13)$$

$$p(x, y, 0) = p_0(x, y), \quad (9.14)$$

$$\phi(x, y, 0) = \phi_0(x, y), \quad (9.15)$$

where ϕ_0 is obtained by solving Poisson's equation with the initial dopants, defects, and carrier concentrations. Radiation then enters through $\dot{q}_{\text{rad}}$, S_j , G_{rad} , and $\dot{q}_{\text{th}}$.

10 Time scales and stiffness

Module 6 contains processes with very different characteristic times. Carrier redistribution can be extremely fast compared with defect annealing. Thermal diffusion may be intermediate, but ballistic heat-carrier effects can introduce nonlocal transport on short time and length scales. Defect reactions may persist long after the initial radiation event.

A useful set of process time scales is

$$\tau_{\text{car}} \sim \frac{L^2}{D_n} \quad \text{or} \quad \frac{L}{\mu_n |\mathbf{E}|}, \quad (10.1)$$

$$\tau_{\text{th}} \sim \frac{L^2}{\alpha}, \quad \alpha = \frac{k}{\rho_m c_p}, \quad (10.2)$$

$$\tau_{\text{def},j} \sim \frac{L^2}{D_j}, \quad (10.3)$$

$$\tau_{\text{ann},j} \sim \frac{1}{k_{\text{ann},j}}, \quad (10.4)$$

$$\tau_{\text{rec}} \sim \frac{1}{\sigma v_{\text{th}} N_t}. \quad (10.5)$$

These time scales can differ by many orders of magnitude. Therefore, the coupled solver should not assume that one universal time step resolves all physics equally well. In early MATLAB implementation, it is better to start with quasi-static subproblems and then introduce time dependence gradually.

10.1 Dimensionless indicators

Several dimensionless groups help decide which couplings matter.

The defect Damkohler number compares transport and reaction:

$$\text{Da}_j = \frac{L^2/D_j}{1/k_j} = \frac{k_j L^2}{D_j}. \quad (10.6)$$

If $\text{Da}_j \ll 1$, diffusion is fast compared with reaction. If $\text{Da}_j \gg 1$, reaction is fast compared with diffusion.

A carrier Peclet number compares drift and diffusion:

$$\text{Pe}_n = \frac{\mu_n |\mathbf{E}| L}{D_n}, \quad \text{Pe}_p = \frac{\mu_p |\mathbf{E}| L}{D_p}. \quad (10.7)$$

If $\text{Pe} \gg 1$, drift dominates; if $\text{Pe} \ll 1$, diffusion dominates.

For Module 4, the Knudsen number compares heat-carrier mean free path with device length:

$$\text{Kn} = \frac{\Lambda}{L}. \quad (10.8)$$

If $\text{Kn} \ll 1$, Fourier-like diffusion is more accurate. If $\text{Kn} \sim 1$, ballistic-diffusive physics matters.

The Biot number compares boundary heat transfer with internal conduction:

$$\text{Bi} = \frac{hL}{k}. \quad (10.9)$$

Large Bi means the boundary can strongly influence the internal temperature profile.

11 Recommended first implementation path

A full Module 6 implementation can become complicated quickly. The first version should preserve the coupling architecture while enabling verification at each level. A practical staged path is as follows.

11.1 Level 0: one-way coupling

Run each module once in a physically ordered sequence:

$$\dot{q}_{\text{rad}} \rightarrow \mathbf{C} \rightarrow \phi, \mathbf{E} \rightarrow T \rightarrow n, p, \mathbf{J} \rightarrow y_{\text{device}}. \quad (11.1)$$

No feedback iteration is applied. This tests data formats, interpolation, units, and field alignment.

11.2 Level 1: electrostatic-carrier coupling

Couple Poisson and drift-diffusion while holding $\mathbf{C}$ and T fixed:

$$(n, p) \leftrightarrow \rho \leftrightarrow \phi, \mathbf{E}. \quad (11.2)$$

This is the classical semiconductor-device coupling and is the most important step for current-voltage response.

11.3 Level 2: defect-electrostatic-carrier coupling

Allow the defect charge and trap density to affect both electrostatics and recombination:

$$\mathbf{C} \rightarrow \rho_{\text{def}}, N_t \rightarrow \phi, \mathbf{E}, R_{\text{SRH}} \rightarrow n, p, \mathbf{J}. \quad (11.3)$$

At this level, radiation damage directly modifies the device electrical response.

11.4 Level 3: thermal feedback

Add temperature-dependent coefficients and heat generation:

$$T \leftrightarrow (D_j, k_{\text{ann},j}, \mu_n, \mu_p, R), \quad \mathbf{J} \cdot \mathbf{E} \rightarrow T. \quad (11.4)$$

This level captures self-heating, thermally enhanced defect kinetics, and temperature-dependent transport degradation.

11.5 Level 4: full coupled iteration

Iterate all maps until convergence at every time step:

$$\mathbf{C} \leftrightarrow \phi, \mathbf{E} \leftrightarrow n, p, \mathbf{J} \leftrightarrow T \leftrightarrow \mathbf{C}. \quad (11.5)$$

This is the long-term target of Module 6.

12 Conservation and consistency checks

Module 6 must be checked more carefully than a single-physics module because coupling errors can produce apparently reasonable but physically wrong fields.

12.1 Charge consistency

At every coupled iteration, the charge density used in Poisson's equation should match the current defect and carrier fields:

$$\rho_{\text{check}} = q \left[p - n + N_D^+ - N_A^- + \sum_j z_j C_j \right]. \quad (12.1)$$

A useful diagnostic is

$$\epsilon_\rho = \frac{\|\rho_{\text{Poisson}} - \rho_{\text{check}}\|_2}{\|\rho_{\text{check}}\|_2 + \rho_{\text{floor}}}. \quad (12.2)$$

This should be close to zero if the coupling is implemented consistently.

12.2 Energy consistency

The thermal source should be decomposed and tracked:

$$Q_{\text{tot}} = \dot{q}_{\text{th}} + Q_{\text{J}} + Q_{\text{rec}} + Q_{\text{def}} + Q_{\text{ext}}. \quad (12.3)$$

A useful output is the integrated power in each channel:

$$P_a(t) = \int_{\Omega} Q_a(x, y, t) d\Omega, \quad (12.4)$$

where $a \in \{\text{th}, \text{J}, \text{rec}, \text{def}, \text{ext}\}$. This reveals whether one heat source dominates and whether the magnitude is physically plausible.

12.3 Carrier conservation

For an insulating device domain with no carrier flux through the boundary, the total number of carriers should change only through generation and recombination:

$$\frac{d}{dt} \int_{\Omega} n d\Omega = \int_{\Omega} (G - R) d\Omega, \quad (12.5)$$

$$\frac{d}{dt} \int_{\Omega} p d\Omega = \int_{\Omega} (G - R) d\Omega. \quad (12.6)$$

With contacts, boundary current terms must be included. For electrons,

$$\frac{d}{dt} \int_{\Omega} n d\Omega = \frac{1}{q} \int_{\partial\Omega} \mathbf{J}_n \cdot \mathbf{n} d\Gamma + \int_{\Omega} (G - R) d\Omega. \quad (12.7)$$

For holes,

$$\frac{d}{dt} \int_{\Omega} p d\Omega = -\frac{1}{q} \int_{\partial\Omega} \mathbf{J}_p \cdot \mathbf{n} d\Gamma + \int_{\Omega} (G - R) d\Omega. \quad (12.8)$$

These identities are important sign checks for Module 5 inside Module 6.

13 Device-response outputs

The final output of Module 6 is not merely a set of fields. The project goal is to connect radiation damage to measurable device degradation. Typical outputs include:

$$I_a(t) = \int_{\Gamma_a} \mathbf{J} \cdot \mathbf{n}_a d\Gamma, \quad (13.1)$$

$$\Delta I_{\text{leak}}(t) = I_{\text{leak}}(t) - I_{\text{leak},0}, \quad (13.2)$$

$$\Delta V_{\text{th}}(t) = V_{\text{th}}(t) - V_{\text{th},0}, \quad (13.3)$$

$$Q_{\text{col}}(t) = \int_0^t \int_{\Gamma_a} \mathbf{J} \cdot \mathbf{n}_a d\Gamma dt', \quad (13.4)$$

$$y_{\text{device}} = \mathcal{G}(\mathbf{C}, T, \phi, n, p, \mathbf{J}). \quad (13.5)$$

Here I_a is terminal current through contact a , ΔI_{leak} is leakage-current shift, ΔV_{th} is threshold-voltage shift, Q_{col} is collected charge, and $\mathcal{G}$ is a reduced post-processing map from fields to device-level response.

For the first project implementation, a simple degradation score can be constructed from normalized metrics:

$$S_{\text{deg}} = w_C \frac{\|\mathbf{C} - \mathbf{C}_0\|}{\|\mathbf{C}_0\| + C_{\text{floor}}} + w_T \frac{\|T - T_0\|}{\|T_0\|} + w_I \frac{|I - I_0|}{|I_0| + I_{\text{floor}}} + w_R \frac{\int_{\Omega} R d\Omega}{R_{\text{ref}}}. \quad (13.6)$$

This is not a fundamental physics law. It is a reduced scoring model that can help compare simulations and decide which physics channels dominate the final response.

14 Common failure modes

14.1 Unit inconsistency

A frequent error is mixing volumetric sources and area-integrated sources. If $\dot{q}_{\text{rad}}$ is a volumetric rate, its units are W/m³. If it has been depth-integrated rather than depth-averaged, its units become W/m². Module 6 should use a depth-averaged volumetric source when the two-dimensional

model represents an out-of-plane averaged material volume.

14.2 Using stale fields

If the defect map is updated but Poisson's equation still uses the previous defect map, the potential is inconsistent. If the current density is updated but the thermal source still uses the previous current density, the thermal field is inconsistent. Module 6 should explicitly record which time level and coupling iteration each field comes from.

14.3 Sign errors in carrier transport

Electron particles drift opposite $\mathbf{E}$; holes drift along $\mathbf{E}$. Conventional electron current points opposite electron particle motion. This distinction must be preserved in code, notes, and figures.

14.4 Over-coupling too early

Turning on every feedback loop at once can make debugging impossible. The recommended approach is to verify one-way coupling, then electrostatic-carrier coupling, then defect-electrostatic-carrier coupling, and only then thermal feedback.

15 Summary

Module 6 is the self-consistent coupling layer for the radiation-effects project. It joins the Module 3 defect map, Module 2 electrostatic potential map, Module 4 temperature map, and Module 5 carrier map over the same two-dimensional device cross section. The essential coupling equations are

$$\rho = q \left[p - n + N_D^+ - N_A^- + \sum_j z_j C_j \right], \quad (15.1)$$

$$\mathbf{E} = -\nabla_{xy} \phi, \quad (15.2)$$

$$\mathbf{J}_n = q\mu_n n \mathbf{E} + qD_n \nabla_{xy} n, \quad (15.3)$$

$$\mathbf{J}_p = q\mu_p p \mathbf{E} - qD_p \nabla_{xy} p, \quad (15.4)$$

$$Q_{\text{tot}} = \dot{q}_{\text{th}} + \mathbf{J} \cdot \mathbf{E} + Q_{\text{rec}} + Q_{\text{def}} + Q_{\text{ext}}. \quad (15.5)$$

These equations express the key feedbacks: defects change charge density, mobility, and recombination; carriers change space charge and heat generation; temperature changes mobilities and defect kinetics; and the potential steers carrier and charged-defect motion.

The first implementation should not begin with a fully monolithic solve. It should begin with one-way map passing, then add fixed-point coupling between selected modules. Once the field maps

converge under iteration, Module 6 becomes the physics bridge from radiation-induced material damage to measurable device response.